

Product Requirements Document (PRD)
Algorithmic Trading System

Document Control

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1. Executive Summary

The Algorithmic Trading System (ATS) is a strategic initiative to develop a high-performance, user-friendly platform for simulated trading across multiple asset classes, including stocks, indices, forex, and commodities, on major global exchanges such as NYSE, NASDAQ, LSE, and NSE. Integrated with the Interactive Brokers (IBKR) Paper Trading API, the ATS leverages advanced data processing libraries (Polars, DuckDB) and AI agents to enable traders to test and refine trading strategies in a risk-free environment. The system features two core trading strategies—Trend-Following and Support and Resistance—along with a responsive Market Watch Dashboard for real-time market monitoring, technical analysis, portfolio management, and simulated trading.

The ATS addresses the hedge fund's need to modernize trading operations by automating processes, enhancing decision-making, and ensuring robust risk management. It aims to deliver competitive advantages through faster trade execution, data-driven insights, and compliance with financial regulations. Key features include automated trading strategies, real-time data visualization, AI-driven recommendations, and comprehensive reporting. The system is designed for scalability, with a modular architecture that supports future enhancements such as live trading and alternative data integration.

This Product Requirements Document (PRD) provides a detailed blueprint for the ATS, outlining functional and non-functional requirements, integration needs, and implementation considerations. It serves as a guiding document for stakeholders, including product managers, developers, traders, and compliance officers, to align on the product's vision, scope, and deliverables. The ATS will be developed using Agile Scrum with Test-Driven Development (TDD) and Continuous Integration/Continuous Deployment (CI/CD) practices, targeting a 3-6 month timeline for initial deployment on a local Lenovo Legion Pro 5 system.

The ATS is expected to drive operational efficiency, reduce risks, and provide a foundation for future growth, positioning the hedge fund as a leader in algorithmic trading. Success will be measured by user adoption, strategy performance, system uptime, and compliance adherence, with a focus on delivering value to traders and stakeholders.

2. Introduction

2.1. Purpose

The purpose of this PRD is to define the product requirements for the Algorithmic Trading System, a critical platform for the hedge fund's trading operations. This document articulates the product's vision, objectives, and detailed requirements to guide the development team in building a robust, scalable, and user-friendly system. It serves as a contract between stakeholders—product management, development, trading, and compliance teams—to ensure alignment on the system's functionality, performance, and strategic goals. The PRD provides a comprehensive framework for designing, implementing, and validating the ATS, ensuring it meets the hedge fund's business needs and regulatory standards.

2.2. Scope

The ATS is a comprehensive platform for algorithmic paper trading, encompassing the following components:

- **Trading Strategies:** Two positional strategies—Trend-Following and Support and Resistance—for long and short positions across stocks, indices, forex, and commodities.
- **Market Watch Dashboard:** A web-based, responsive interface for real-time market monitoring, technical analysis, portfolio management, and simulated trading.
- **Data Management:** Ingestion, processing, storage, and visualization of market data using Polars and DuckDB.
- **Order Execution:** Placement and monitoring of simulated trades via the IBKR Paper Trading API.
- **Risk Management:** Position sizing, stop-loss, take-profit, and portfolio-level risk controls.
- **AI Agents:** Modular components for market analysis, trade execution, and risk assessment, operating under human oversight.
- **Reporting and Analytics:** Trade performance metrics, backtesting results, portfolio analytics, and compliance reports.

- **Security and Compliance:** Encrypted communications, secure credential management, and adherence to financial regulations.

Out of Scope:

- Live trading functionality (planned for future phases).
- Integration with brokers other than IBKR.
- Support for cryptocurrencies or non-traditional asset classes.
- Mobile app development (dashboard will be mobile-responsive).
- Advanced AI model training (pre-trained models will be used).
- Premium third-party data sources (e.g., Bloomberg, Reuters) for initial phase.

2.3. Objectives

The ATS aims to achieve the following product objectives:

1. **Automate Trading Processes:** Enable rapid, accurate execution of algorithmic strategies to reduce manual effort and improve efficiency.
2. **Enhance Decision-Making:** Provide real-time market data, technical indicators, and AI-driven insights to support informed trading decisions.
3. **Mitigate Financial Risks:** Implement robust risk management protocols to protect capital and ensure sustainable returns.
4. **Facilitate Strategy Validation:** Support backtesting and paper trading to test strategies without financial risk.
5. **Deliver Intuitive User Experience:** Create a responsive, user-friendly Market Watch Dashboard to maximize trader productivity.
6. **Ensure Regulatory Compliance:** Adhere to financial regulations (e.g., SEC, GDPR, MiFID II, SEBI) through audit trails and reporting.
7. **Build for Scalability:** Design a modular system to support future enhancements, such as live trading and alternative data integration.
8. **Drive Competitive Advantage:** Position the hedge fund as a leader in algorithmic trading through advanced technology and data-driven strategies.

2.4. Assumptions and Constraints

Assumptions:

- The IBKR Paper Trading API provides reliable, timely market data and order execution capabilities.
- Traders have intermediate knowledge of algorithmic trading and technical analysis.
- The Lenovo Legion Pro 5 (NVIDIA RTX 3060 GPU) meets performance requirements for development and testing.
- Regulatory requirements for paper trading are less stringent than for live trading.
- The development team has expertise in Python, Polars, DuckDB, React, and FastAPI.
- Budget and resources are sufficient for a 3-6 month development cycle.

Constraints:

- Development timeline: 3-6 months to align with strategic goals.
- IBKR API rate limits: 50 requests/second for market data, 100 messages/second for orders.
- Budget: No premium third-party data sources or cloud hosting for initial deployment.
- Deployment: Local on Lenovo Legion Pro 5, no cloud infrastructure.
- Compliance: Must adhere to SEC, GDPR, MiFID II, and SEBI regulations.
- Hardware: Limited to existing infrastructure (Lenovo Legion Pro 5).

2.5. Stakeholders

Stakeholder	Role	Responsibilities
Head of Trading	Oversees trading operations	Defines strategy requirements, validates functionality
Traders	Primary end-users	Test system, provide usability feedback
Product Owner	Represents business interests	Prioritizes features, manages backlog
Development Team	Builds and tests system	Implements requirements, ensures quality
Compliance Officer	Ensures regulatory adherence	Reviews system for compliance
IT Administrator	Manages infrastructure	Configures hardware, ensures uptime
Risk Manager	Oversees risk controls	Defines risk parameters, monitors performance
Executive Leadership	Approves strategic initiatives	Allocates resources, reviews progress

2.6. Definitions and Acronyms

- **ATS:** Algorithmic Trading System.
 - **IBKR:** Interactive Brokers.
 - **OHLCV:** Open, High, Low, Close, Volume.
 - **VW SMA:** Volume-Weighted Simple Moving Average.
 - **MFI:** Money Flow Index.
 - **ATR:** Average True Range.
 - **Polars:** High-performance data processing library.
 - **DuckDB:** In-memory analytical database.
 - **FastAPI:** Python framework for building APIs.
 - **ib_insync:** Python library for IBKR API integration.
 - **Gann Fan:** Technical analysis tool for dynamic support/resistance.
 - **Fibonacci Retracement:** Percentage-based retracement levels.
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3. Business Context

3.1. Business Need

The hedge fund seeks to modernize its trading operations to remain competitive in a rapidly evolving financial landscape. The ATS addresses the following business needs:

- **Operational Efficiency:** Automating trading processes reduces manual effort, minimizes errors, and accelerates execution, enabling traders to focus on strategy development.
- **Competitive Advantage:** Algorithmic trading provides faster, more precise trade execution compared to manual methods, positioning the fund to outperform competitors.
- **Risk Management:** Robust risk controls are essential to protect capital in volatile markets, ensuring sustainable returns and investor confidence.
- **Strategy Validation:** Paper trading and backtesting allow the fund to test strategies without financial risk, improving confidence in deployment.
- **Data-Driven Decisions:** Real-time market data, technical analysis, and AI-driven insights empower traders to capitalize on opportunities.
- **Regulatory Compliance:** A compliant system mitigates risks of penalties and reputational damage, critical in a heavily regulated industry.
- **Scalability:** A modular platform supports future growth, such as live trading and alternative data integration, aligning with long-term strategic goals.

3.2. Market Opportunity

The global algorithmic trading market is projected to grow at a CAGR of 12-15% through 2030, driven by technological advancements, increasing market volatility, and the adoption of AI and machine learning in finance. Key market trends include:

- **Rise of Algorithmic Trading:** Both institutional and retail investors are adopting algorithmic trading for its efficiency, scalability, and ability to exploit market inefficiencies.
- **Multi-Asset Platforms:** Demand for systems supporting diverse asset classes (stocks, forex, commodities) is increasing to diversify portfolios and manage risk.
- **AI and Machine Learning:** AI-driven trading systems are gaining traction for their ability to analyze complex datasets, predict trends, and adapt to market changes.
- **Regulatory Evolution:** Stricter regulations (e.g., MiFID II, SEC) are driving demand for compliant, transparent trading systems with robust audit trails.
- **Technology Adoption:** High-performance computing, advanced data analytics (e.g., Polars, DuckDB), and cloud infrastructure are becoming standard in trading platforms.

The ATS positions the hedge fund to capitalize on these trends by offering a multi-asset, AI-augmented platform with strong compliance and performance capabilities. By focusing on paper trading initially, the system allows the fund to refine strategies and build a foundation for live trading, addressing both immediate and long-term market opportunities.

3.3. Competitive Landscape

The algorithmic trading market includes a mix of proprietary platforms, commercial software, and open-source solutions. Key competitors include:

- **Proprietary Systems:**

- **Goldman Sachs, JPMorgan:** In-house platforms with advanced AI and high-frequency trading capabilities. Strengths: Customization, performance. Weaknesses: High cost, limited accessibility.
- **Commercial Platforms:**
 - **MetaTrader 5:** Popular for forex and stocks, with scripting and backtesting. Strengths: User-friendly, widely adopted. Weaknesses: Limited AI integration, less flexible for custom strategies.
 - **TradeStation:** Offers advanced charting and automation. Strengths: Robust analytics, community support. Weaknesses: High licensing costs, complex for beginners.
- **Open-Source Solutions:**
 - **QuantConnect:** Cloud-based platform with extensive backtesting. Strengths: Free, community-driven. Weaknesses: Limited real-time execution, steep learning curve.
 - **Backtrader:** Python-based framework for strategy development. Strengths: Flexible, lightweight. Weaknesses: Lacks integrated UI, requires custom integrations.

ATS Differentiation:

- **Custom Strategies:** Tailored Trend-Following and Support and Resistance strategies for positional trading, optimized for the hedge fund's risk profile.
- **AI Integration:** Modular AI agents for analysis, execution, and risk management, providing a competitive edge over platforms with limited AI capabilities.
- **User Experience:** A responsive Market Watch Dashboard designed for traders, balancing simplicity and advanced functionality.
- **Compliance Focus:** Built-in audit trails and reporting to meet SEC, GDPR, MiFID II, and SEBI requirements, addressing regulatory gaps in open-source solutions.
- **Local Deployment:** Leverages existing hardware (Lenovo Legion Pro 5) to reduce costs, unlike cloud-based competitors with ongoing fees.
- **Scalability:** Modular architecture supports future live trading and data integration, unlike rigid commercial platforms.

3.4. Regulatory Considerations

The ATS must comply with financial regulations, even in paper trading mode, to prepare for future live trading. Key regulations include:

- **SEC (U.S.):** Requires transaction reporting, market abuse prevention, and data protection under Regulation NMS and CAT.
- **GDPR (Europe):** Mandates protection of personal and financial data, with explicit consent for processing.
- **MiFID II (Europe):** Enforces transparency in trade execution and reporting for EU markets.
- **SEBI (India):** Governs algorithmic trading on NSE, requiring risk controls and audit trails.
- **Data Security:** Adherence to ISO 27001 and PCI DSS for secure data handling.

The ATS will implement audit trails, transaction logging, and compliance monitoring to meet these requirements. The Compliance Officer will review all features to ensure regulatory alignment, with a focus on preparing the system for live trading compliance.

4. Product Overview

4.1. Product Description

The Algorithmic Trading System (ATS) is a high-performance platform for simulated trading, enabling traders to execute, monitor, and analyze algorithmic strategies in a risk-free environment. Integrated with the IBKR Paper Trading API, the ATS supports trading across stocks, indices, forex, and commodities on major exchanges. It features two positional trading strategies (Trend-Following and Support and Resistance), a responsive Market Watch Dashboard, and AI agents for enhanced decision-making. The system uses Polars and DuckDB for efficient data processing and storage, with a modular, event-driven architecture to ensure scalability and maintainability. Deployed locally on the Lenovo Legion Pro 5, the ATS prioritizes performance, security, and compliance, providing traders with a powerful tool to test strategies and optimize performance.

4.2. Vision and Goals

Vision: To empower the hedge fund with a cutting-edge algorithmic trading platform that delivers automated, data-driven trading capabilities, robust risk management, and regulatory compliance, positioning the fund as a leader in financial innovation.

Goals:

- 1. **Short-Term (3-6 Months):**
 - Deliver a fully functional paper trading system with two trading strategies.
 - Achieve 100% trader adoption for the Market Watch Dashboard.
 - Ensure system uptime of 99.9% and compliance with regulatory requirements.
- 2. **Medium-Term (6-12 Months):**
 - Validate strategies through backtesting and paper trading, achieving a Sharpe ratio > 1.5 in simulations.
 - Expand system capabilities to include additional asset classes or strategies.
 - Prepare for live trading integration with minimal architectural changes.
- 3. **Long-Term (12+ Months):**
 - Transition to live trading with seamless integration of real capital.
 - Integrate alternative data sources (e.g., news, sentiment) for enhanced decision-making.
 - Scale the system to support cloud deployment and larger trading volumes.

4.3. Key Features

- **Multi-Asset Trading:** Supports stocks (e.g., AAPL, RELIANCE), indices (e.g., S&P 500, NIFTY 50), forex (e.g., EUR/USD), and commodities (e.g., gold, crude oil).
- **Automated Strategies:** Executes Trend-Following and Support and Resistance strategies with dynamic position sizing and risk controls.
- **Real-Time Monitoring:** Displays live prices, volume, and technical indicators via the Market Watch Dashboard.
- **Interactive Dashboard:** Offers candlestick charts, portfolio summaries, and simulated trading controls, optimized for desktop and mobile.
- **AI-Driven Insights:** Provides market analysis, trade execution, and risk assessment through modular AI agents.
- **Backtesting and Simulation:** Validates strategies against historical data and simulates trades in the IBKR paper trading environment.
- **Robust Security:** Implements encrypted communications, secure credential management, and role-based access control.
- **Comprehensive Reporting:** Generates trade performance metrics, portfolio analytics, and compliance reports.

4.4. User Personas

Persona	Description	Goals	Pain Points
Lead Trader	Experienced managing portfolios.	Optimize strategies, monitor performance.	Manual processes, lack of real-time insights.
Junior Trader	New to algorithmic trading, tech-savvy.	Learn strategies, execute trades easily.	Complex platforms, steep learning curve.
Compliance Officer	Ensures regulatory adherence.	Verify compliance, generate reports.	Incomplete audit trails, regulatory risks.
IT Administrator	Manages system infrastructure.	Ensure uptime, secure deployment.	Resource constraints, complex integrations.

4.5. Success Metrics

- **User Adoption:** 100% of traders using the Market Watch Dashboard within 1 month of launch.
 - **Strategy Performance:** Achieve Sharpe ratio > 1.5 and win rate > 60% in paper trading simulations.
 - **System Performance:** End-to-end latency <100ms, uptime 99.9%.
 - **Usability:** Average time to complete key tasks (e.g., place trade, view portfolio) <30 seconds.
 - **Compliance:** Zero regulatory violations, 100% audit trail coverage.
 - **Backtesting Accuracy:** Backtest results within 5% of paper trading outcomes.
 - **Trader Satisfaction:** Net Promoter Score (NPS) > 70 based on post-launch survey.
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5. Functional Requirements

5.1. Trading Strategies

The ATS implements two positional trading strategies: Trend-Following and Support and Resistance, supporting long and short positions across multiple asset classes.

5.1.1. Trend-Following Strategy

Description: A strategy that captures long-term price trends using volume-weighted technical indicators to assess direction, momentum, and volatility.

Requirements:

- **Assets:** Stocks (NYSE, NASDAQ, LSE, NSE), indices (S&P 500, FTSE 100, NIFTY 50), forex (EUR/USD, USD/JPY), commodities (gold, crude oil).
- **Technical Indicators:**
 - 55-day VW SMA of Open: Identifies long-term trend.
 - Formula:
$$\left[\text{VW SMA} = \frac{\sum (\text{Open Price}_i \times \text{Volume}_i)}{\sum \text{Volume}_i} \right]$$
 - 13-day VW SMA of Open: Tracks medium-term trends (optional).
 - 14-day VW MFI of HLC Average: Measures momentum.
 - HLC Average = (High + Low + Close) / 3.
 - Raw Money Flow = HLC Average × Volume.
 - MFI = 100 - (100 / (1 + (Positive Money Flow / Negative Money Flow))).
 - 34-day VW SMA of MFI(14): Smooths MFI.
 - 21-day VW SMA of MFI(14): Triggers exits.
 - 5-day VW SMA of High: Short-term resistance.
 - 5-day VW SMA of Low: Short-term support.
 - 21-day VW ATR: Volatility measure.
 - True Range = max(High - Low, |High - Previous Close|, |Low - Previous Close|).
 - ATR = VW SMA of True Range.
- **Entry Rules:**
 - Long: 14-day VW MFI > 34-day VW SMA of MFI(14) and price > 55-day VW SMA of Open.
 - Short: 14-day VW MFI < 34-day VW SMA of MFI(14) and price < 55-day VW SMA of Open.
- **Exit Rules:**
 - Long:
 - Stop-Loss: Entry price - (2 × 21-day VW ATR).
 - Take-Profit: Entry price + (2 × 21-day VW ATR).
 - Additional: MFI(14) < 21-day VW SMA of MFI(14) or price < 5-day VW SMA of Low.
 - Short:
 - Stop-Loss: Entry price + (2 × 21-day VW ATR).
 - Take-Profit: Entry price - (2 × 21-day VW ATR).
 - Additional: MFI(14) > 21-day VW SMA of MFI(14) or price > 5-day VW SMA of High.
- **Position Sizing:**
 - Risk: 1% of account balance (£10,000 initial).
 - Formula:
$$\left[\text{Position Size} = \frac{\text{Account Balance} \times 0.01}{2 \times \text{21-day VW ATR}} \times \text{Lot Size} \right]$$
 - Lot Size: Stocks (1 share), forex (100,000 units), commodities (per contract), indices (per multiplier).
- **Money Management:**

- Drawdown > 20%: Reduce risk to 0.5%.
- Trailing Stop: $1.5 \times 21\text{-day VW ATR}$.
- Convert to GBP using IBKR exchange rates.

5.1.2. Support and Resistance Strategy

Description: Trades breakouts and reversals at key support/resistance levels using Pivot Points, Fibonacci retracement, and Gann Fan.

Requirements:

- **Assets:** Same as Trend-Following.
- **Technical Tools:**
 - Pivot Points:
 - $PP = (High + Low + Close) / 3$.
 - $R1 = 2 \times PP - Low$.
 - $S1 = 2 \times PP - High$.
 - $R2 = PP + (High - Low)$.
 - $S2 = PP - (High - Low)$.
 - Fibonacci Retracement: 23.6%, 38.2%, 50%, 61.8%, 100%.
 - Gann Fan: 1x1, 1x2, 2x1 angles from highs/lows.
- **Entry Rules:**
 - Long: Price breaks above R1 or Fibonacci level (e.g., 61.8%) with rising volume; or bounces off Gann Fan angle in uptrend.
 - Short: Price breaks below S1 or Fibonacci level (e.g., 38.2%) with rising volume; or rejected at Gann Fan angle in downtrend.
 - Optional: Confirm with candlestick patterns (e.g., bullish engulfing, bearish shooting star).
- **Exit Rules:**
 - Long:
 - Take-Profit: R2, next Fibonacci, or Gann Fan angle.
 - Stop-Loss: Below S1, swing low, or 1-2% below entry.
 - Short:
 - Take-Profit: S2, next Fibonacci, or Gann Fan angle.
 - Stop-Loss: Above R1, swing high, or 1-2% above entry.
 - Trailing Stop: 2-3% or $2 \times 14\text{-day ATR}$.
- **Position Sizing:**
 - Risk: 1-2% of account balance.
 - Formula:

$$\left[\frac{\text{Account Balance} \times \text{Risk \%}}{|\text{Entry Price} - \text{Stop Loss Price}|} \right]$$
- **Money Management:**
 - Portfolio risk: <5-10%.
 - Diversify across assets/sectors.
 - Adjust position sizes using 14-day ATR.

5.2. Market Watch Dashboard

Description: A responsive web interface for real-time market monitoring, technical analysis, portfolio management, and simulated trading.

5.2.1. Scrip Selection

- Dropdown/searchable input for tickers (e.g., AAPL, EUR/USD).
- Support multiple exchanges.
- Auto-suggest tickers; validate via IBKR API.

5.2.2. Real-Time Data Display

- Show bid, ask, last price, volume, OHLCV.
- Update every 5-10 seconds via WebSocket.
- Highlight price changes (green up, red down).
- Fields: Ticker, exchange, bid, ask, last, volume, % change.

5.2.3. Interactive Charts

- Candlestick/line charts with zoom/pan.
- Overlay indicators: VW SMA, MFI, ATR, RSI(14), MACD, Pivot Points, Fibonacci, Gann Fan.
- Timeframes: 1m, 5m, 15m, 1h, 4h, 1d, 1w.
- Features: Crosshair, export PNG, save settings.

5.2.4. Portfolio Summary

- Display positions, account value, unrealized P&L.
- Fields: Ticker, size, entry price, current price, P&L, holding period.
- Filter by asset, exchange, strategy.

5.2.5. Simulated Trading Panel

- Place buy/sell orders (market, limit, stop, bracket).
- Input: Ticker, quantity, price, order type.
- Validate inputs; show order status.
- Features: Preview impact, cancel/modify orders.

5.2.6. Watchlist

- Save favorite scrips.
- Highlight high volatility/volume scrips.
- Reorder/group by asset type.

5.2.7. Technical Analysis

- Daily/weekly outlook: Strong Uptrend, Uptrend, Sideways, Downtrend, Strong Downtrend.
- Indicator status: RSI(14), MFI(14), MACD, VW SMA, EMA.
- Support/resistance levels with AI recommendations.

5.2.8. News and Events

- Show scrip-specific news and events (AGM, dividends, earnings) within 10 days.
- Filter by relevance/date; highlight high-impact events.
- Placeholder for future API integration.

5.3. Data Management

Description: Manages market data for trading and analysis.

5.3.1. Data Ingestion

- Fetch real-time OHLCV via IBKR WebSocket.
- Retrieve historical data (5 years) for backtesting.
- Poll every 5-10 seconds; cache in Redis.

5.3.2. Data Processing

- Normalize timestamps (UTC), align asset IDs.
- Clean data: Remove outliers, interpolate missing values.
- Compute indicators using Polars.

5.3.3. Data Storage

- Store in DuckDB with Parquet, partitioned by date/asset.
- Retain 7 years for compliance.
- Daily backups to external storage.

5.3.4. Data Visualization

- Render price/indicator/performance charts.
- Support multi-timeframe analysis.
- Export as PNG/CSV.

5.4. Order Execution

Description: Manages trade lifecycle in IBKR paper trading.

5.4.1. Order Types

- Market, limit, stop, bracket (entry, take-profit, stop-loss).
- Support partial fills.

5.4.2. Order Placement

- Submit via `ib_insync` with validation.
- Calculate position size; confirm submission.

5.4.3. Order Monitoring

- Track status (pending, filled, canceled).
- Update portfolio/P&L; notify users.

5.4.4. Error Handling

- Handle API errors; retry 3 times with backoff.
- Log errors; alert for critical failures.

5.5. Risk Management

Description: Protects capital with robust controls.

5.5.1. Position Sizing

- Risk: 1-2% per trade.
- Trend-Following: Use 21-day VW ATR.
- Support and Resistance: Use stop-loss distance.

5.5.2. Stop-Loss and Take-Profit

- Trend-Following: $2 \times 21\text{-day VW ATR}$.
- Support and Resistance: S1/R1 or 1-2%.
- Place with IBKR API.

5.5.3. Trailing Stop

- Trend-Following: $1.5 \times 21\text{-day VW ATR}$.
- Support and Resistance: 2-3% or $2 \times 14\text{-day ATR}$.
- Update dynamically.

5.5.4. Portfolio Risk

- Limit open risk to 5-10%.
- Monitor correlations; diversify assets.

5.5.5. Drawdown Limits

- Drawdown > 20%: Reduce risk to 0.5%.
- Drawdown > 30%: Halt trading.

5.5.6. Circuit Breakers

- Halt during extreme volatility (>5% price change in 5min).
- Pause for system anomalies; resume after approval.

5.5.7. Currency Handling

- Convert to GBP using IBKR rates.
- Cache rates; log conversions.

5.6. AI Agents Integration

Description: Automates trading tasks with human oversight.

5.6.1. Trade Execution Agent

- Execute trades based on signals.
- Optimize timing/price using reinforcement learning.

5.6.2. Risk Management Agent

- Enforce sizing, exposure, circuit breakers.
- Adjust risk dynamically.

5.6.3. Market Analysis Agent

- Analyze trends, volatility, momentum.
- Generate recommendations using LSTM models.

5.6.4. IBKR API Agent

- Manage API interactions; optimize calls.
- Handle rate limits/errors.

5.6.5. Transparency and Oversight

- Log decisions with rationale.
- Require human approval for critical actions.
- Allow pausing/overriding via dashboard.

5.7. Reporting and Analytics

Description: Generates performance and compliance reports.

5.7.1. Trade Performance

- Metrics: P&L, win rate, Sharpe ratio, max drawdown.
- Visualize equity curves.

5.7.2. Portfolio Analytics

- Exposure, correlations, VaR, expected shortfall.

- Diversification analysis.

5.7.3. Backtesting Reports

- Compare performance vs. benchmarks.
- Include parameter sensitivity.

5.7.4. Compliance Reports

- Transaction logs with timestamps/user IDs.
- Immutable audit trails.

5.7.5. Export Capabilities

- Export as CSV, PDF, PNG.
- Schedule automated reports.

6. Non-Functional Requirements

6.1. Performance

- Latency: <100ms end-to-end.
- Throughput: 10,000 updates/s, 1,000 trades/min.
- Indicator calculation: <1s for 100 scrips.
- Dashboard refresh: Every 5-10s.
- Backtesting: 5 years data in <10min.

6.2. Scalability

- Vertical: Use Lenovo Legion Pro 5 GPU/CPU.
- Horizontal: Support microservices for cloud.
- Data: Handle 1TB historical data.
- Users: 10 concurrent dashboard users.

6.3. Security

- Authentication: OAuth2 with JWT, MFA.
- Data: TLS 1.3, AES-256 encryption.
- Credentials: Encrypted vault.
- API: Rate limiting, CSRF tokens.
- Agents: Sandboxed containers.

6.4. Reliability and Fault Tolerance

- Uptime: 99.9%.
- Error Handling: Auto-reconnect API.
- Data Integrity: Validate market data.
- Circuit Breakers: Halt anomalies.
- Backup: Daily, RTO <1h.

6.5. Usability

- Navigation: <5min training.
- UI Response: <1s.
- Accessibility: WCAG 2.1.
- Documentation: User guides.

6.6. Compliance

- Adhere to SEC, GDPR, MiFID II, SEBI.
- Log trades/orders immutably.
- Retain logs 7 years.
- Monitor market abuse.

7. Integration Requirements

7.1. Interactive Brokers Paper Trading API

- Connection: Host: 127.0.0.1, Port: 7497, Client ID: 0, Account: DUK221396.
- Data: Real-time/historical OHLCV.
- Orders: Market, limit, stop, bracket.
- Rate Limits: 50 req/s (data), 100 msg/s (orders).
- Security: Encrypted credentials.

7.2. External Data Sources

- Placeholder for news (Reuters, Bloomberg).
- Future sentiment/economic data.
- Use REST/WebSocket for real-time.

7.3. Internal Systems

- Sync with portfolio tools.
- Export logs to compliance systems.
- Store data in warehouse via REST.

8. Implementation Considerations

8.1. Development Methodology

- Agile Scrum: 2-week sprints.
- TDD: Write tests first.
- CI/CD: GitHub Actions.

8.2. Technology Stack

- Backend: Python 3.10, FastAPI, ib_insync.
- Data: Polars, DuckDB with Parquet.
- Frontend: React 18, Tailwind CSS, Plotly.js.
- Infrastructure: Docker, GitHub Actions.
- Monitoring: Prometheus, Grafana.

8.3. Timeline and Milestones

- **Month 1:** Requirements, architecture, setup.
- **Month 2:** Data management, strategy engine.
- **Month 3:** Dashboard, order execution.
- **Month 4:** AI agents, risk management.
- **Month 5:** Testing, backtesting.
- **Month 6:** Deployment, user training.

8.4. Resource Requirements

- Team: 2 backend, 2 frontend, 1 AI, 1 DevOps.
- Hardware: Lenovo Legion Pro 5.
- Budget: Internal resources, no external data costs.

9. Risks and Mitigation

Risk	Description		Likelihood	Impact	Mitigation
Market Volatility	Affects performance	strategy	High	High	Diversify assets, circuit breakers
System Failure	Disrupts trading		Medium	High	Auto-reconnect, error handling
Model Errors	AI/strategy errors cause losses		Medium	High	Backtesting, human oversight
Regulatory Compliance	Non-Fines or restrictions		Low	High	Audit trails, compliance reviews
Security Breach	Data or credential leaks		Low	High	Encryption, MFA, security testing
API Limits	Restricts data/orders		Medium	Medium	Caching, optimized calls

10. Assumptions and Dependencies

10.1. Assumptions Log

ID	Description	Impact if Invalid	Validation Plan
A001	IBKR API reliable	Delayed trades	Test API connectivity
A002	Traders know technical analysis	Increased training	Conduct training
A003	Hardware sufficient	Performance issues	Benchmark hardware
A004	Minimal paper trading regulations	Compliance delays	Consult Compliance Officer

10.2. Dependencies

- IBKR API availability.
 - Polars, DuckDB, ib_insync libraries.
 - Lenovo Legion Pro 5 hardware.
 - Team expertise.
-

11. Missing Information and Best Practices

11.1. Missing Information

- Exact VW indicator formulas.
- Detailed trader personas.
- Specific regulatory requirements.
- Performance benchmarks.
- News feed providers.

11.2. Best Practices Applied

- **Indicators:** Standard VW formulas (e.g., $VW\ SMA = \frac{\sum(\text{Price} \times \text{Volume})}{\sum \text{Volume}}$). **Reason:** Industry-standard, ensures accuracy.
 - **Personas:** Assumed intermediate traders. **Reason:** Typical for hedge funds, focuses usability.
 - **Compliance:** Added audit trails. **Reason:** Prepares for live trading, meets common regulations.
 - **Performance:** Targeted <100ms latency. **Reason:** Aligns with high-frequency trading.
 - **News:** Planned Reuters/Bloomberg integration. **Reason:** Reliable financial sources.
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12. Glossary

- **ATS:** Algorithmic Trading System.
 - **IBKR:** Interactive Brokers.
 - **VW SMA:** Volume-Weighted Simple Moving Average.
 - **MFI:** Money Flow Index.
 - **ATR:** Average True Range.
 - **Polars:** Data processing library.
 - **DuckDB:** Analytical database.
 - **FastAPI:** API framework.
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13. Appendices

13.1. User Stories

- As a trader, I want real-time charts so I can monitor price movements.
- As a compliance officer, I want audit trails to ensure regulatory adherence.
- As a junior trader, I want an intuitive dashboard to execute trades easily.

13.2. Mockups and Wireframes

- Dashboard: Grid with scrip selector, chart, portfolio, trading panel.
- Chart: Candlestick with indicators, zoom controls.
- Trading Panel: Form with ticker, quantity, price, order type.

13.3. References

- IBKR API: <https://interactivebrokers.github.io/tws-api/>
- Polars: <https://polars.rs/>
- DuckDB: <https://duckdb.org/>
- FastAPI: <https://fastapi.tiangolo.com/>
- React: <https://reactjs.org/>
- SEC: <https://www.sec.gov/>